Predicting Sports Injury Occurrence Using Machine Learning Methods and Feature Analysis

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Abstract:
Sports injuries are influenced by a combination of physiological, mechanical, environmental, and training-related factors. This study examines how three sets of predictors (25, 18, and 10 variables) impact injury occurrence among athletes using a dataset of approximately 12,000 observations. The variables include physiological measures such as heart rate, body temperature, hydration level, and fatigue index; biomechanical indicators such as joint angles, gait speed, and ground reaction force; and training-related factors including training intensity, duration, and overall load. Environmental conditions, including ambient temperature, humidity, altitude, and playing surface, are also considered, along with demographic variables such as age, gender, body mass index, and sport type. Using statistical modeling techniques, this study evaluates the relationship between these predictors and injury risk to identify the most significant factors. The results aim to improve understanding of injury mechanisms and support the development of data-driven prevention strategies. This study contributes to sports analytics by demonstrating how integrated data can be used to better predict and reduce injury risk in athletes.

Key Words : Injury, Machine Learning, Tree, Sports Analytics, Predictive Modeling, Recall

1. Introduction

Sports injuries are an unavoidable aspect of athletic participation across all levels. Whether in youth, amateur, collegiate, or professional sports, injuries are inevitable. Additionally, all sports have different types of injuries with different timelines of recovery. These can all be detrimental and have lasting impacts on these athletes, as well as fanbases. Injuries in sports can be detrimental for a variety of reasons, with these being performance, cost, mental, and long lasting physical impacts. As a result, injury prevention and prediction have become a critical area of focus within sports science and analytics for the future.

Most athletes train to prevent themselves from becoming injured and to continuously perform their best on the field at all times, but unfortunately these things happen time and time again. With this, when an injury does occur, it can sometimes be detrimental to long-term performance and can derail the trajectory of an athlete's career. Famous athletes like Bo Jackson, Derrick

Rose, and numerous others suffered injuries in the midst of their careers and saw a significant drop in performance after returning.

Financial implications of sports injuries also further highlight their importance. Research published by MDPI shows that injury related absences in professional sports result in monetary losses that are substantial due to medical expenses, rehabilitation costs, and the loss of player ability (Cintia et al., 2021). For example, European soccer teams see significant loss in injury related costs each year, sometimes exceeding over 188 million euros per season. Leagues in the United States are no different, with losses of over 700 million dollars at times in the MLB, and 450 million dollars in the NFL due to injury. These expenses and burdens are not only seen from the organizations too, as sponsors, fans, and the economy also can suffer. Similar findings have been reported in studies, these emphasize the growing financial burden of injuries across sports and competition (Petrovsky et al., 2022). With this, even small improvements can bring in significant economic and competitive advantages in the future.

In addition to the physical and financial impacts that sports injuries can have, there are significant psychological effects that these injuries can have on athletes. Studies in sports medicine include findings from the Journal of Sports Medicine, indicating that injured athletes often show increased levels of stress, anxiety, and decreased motivation during their recovery periods. The pressure to return to prior levels of performance, as well as timelines and pressure, can negatively impact the mental health of high performance athletes. Prolonged injuries also can induce fear, and lack of confidence which further affects performance. These psychological factors help to show that injury impact is not only physical, but instead multifaceted and highlights the importance of early detection and prevention.

With recent increase seen in injuries occurring to athletes on and off the field, it has grown concern among franchises and fans. Looking for effective prevention of injuries helps us to understand the most relevant risk factors that can be found. Even within these, there will be no overall clear consensus, but instead maybe a helping case for what can be prevented. One of the primary challenges for these inquiries is that they are multifactorial. Oftentimes traditional methods have to rely on human eyes, touches, and self assigned fatigue reports with no direct data backing it up in a workload sense. These approaches fail to capture the physiological strain and biomechanical efficiency as well as stress factors (Amendolara et al., 2023). Standard statistical techniques which include logistic regression, often struggle with class imbalance, this then results in models with high accuracy but poor sensitivity. This then fails to identify the athletes at the highest risk for injury as it does not account for injuries being rare events relative to healthy training (Cintia et al., 2021).

Increasing availability of athlete monitoring technologies have further accelerated the potential for data-driven injury prevention. Wearable devices have seen a rise in recent years, with products such as WHOOP and Oura continuously tracking physiological metrics like heart rate variability, sleep, recovery, and overall strain and activity. These technologies allow for large volumes of data to be studied in real time and used to understand injury risk. Recent advancements in artificial intelligence systems have expanded the ability to track health and predict injuries using wearable technology (Gai, 2025). Some research has shown that integrating wearable data with machine learning models can significantly enhance injury prediction and

monitoring opportunities (Tsilimigkras et al., 2024). In this study, a dataset consisting of about 12,000 observations and multiple groups of predictor variables (10, 18, and 25 features) all analyzed and evaluated how different combinations of psychological, biomechanical, and training factors influence injury occurrence. After applying machine learning models to this data, the research seeks to identify patterns and relationships that may not be apparent through other ways of analysis.

This ultimately hopes to contribute to more effective injury detection and prevention. This study aims to utilize machine learning techniques to predict injury occurrence and identify key contributing factors. This demonstrates how modern data collection and analytics can be integrated to improve athlete health, reduce financial loss, and enhance performance by athletes all around the world.

2. Literature Review

Sports injury prediction has become increasingly important to research recently due to the complex and multifactorial aspects of injury occurrence. Nothing is straight forward but data can be used to backtrack many aspects of injuries occurring. Existing publications highlight that injuries are rarely caused by one single factor but instead the interaction of psychological, biomechanical, and training variables (Yuan et al., 2025). As a result of this, research has begun taking along an analytical approach to explore injuries, looking at different methods within machine learning to better understand and help to prevent injury risk.

Early research that helped to predict sports injury prediction relied heavily on traditional models. These approaches helped to identify relationships between individual variables, which could include fatigue, load, or injury history (Cintia et al., 2021). Some studies published regarding sports medicine emphasize that while these methods are easy to interpret at times, they can struggle to capture nonlinear and multifactorial relationships in injury data (Yuan et al., 2025). As an example of this, past research has shown that workload and fatigue are often strongly associated with injury risk, especially when training intensity suddenly increases (Jauhiainen et al., 2021). With these models though, there is often independence assumed between variables and thus they are limited in their ability to account for complex cases. Some models can have good accuracy, but do not always identify high risk athletes, especially in cases where injuries are not very common.

More recent studies show that the use of machine learning techniques are being used to address limitations of prior methods. These methods include trees, support vector machines (SVM), Naive Bayes, and neutral networks. These methods have been better shown to handle nonlinear relationships and datasets with more features and higher dimensions (Amendolara et al., 2023). On top of this, studies have shown the effectiveness of these approaches in predicting injury risk across different sports (Robles-Palazón et al., 2023).

Studies done including those in Cureus and ScienceDirect give comprehensive overviews of machine learning applications in sports injury prevention. These studies help to show that machine learning models can help to vastly improve predictive performance by seeing patterns that are not attainable through statistical approaches (Amendolara et al., 2023; Chu et al., 2022). These machine learning models are especially effective in handling larger datasets with large

predictor numbers. This allows them to be well prepared for injury prediction where multiple contributing factors are involved (Leckey et al, 2025). Research has also shown that machine learning models have better adaptability to different types of data, including physiological metrics, biomechanical indicators, and environmental conditions (Jauhiainen et al, 2022). These models being more flexible allows for more comprehensive modeling of injury risk which is essential when it comes to athletic performance and recovery. This flexibility has been supported by research that demonstrates the ability of machine learning to adapt to complex physiological and biomechanical datasets (Maghsoomi et al., 2025).

A commonly seen theme across literature is the importance of feature selection in improving model performance. Methods which include Minimum Redundancy Maximum relevance (MRMR), are commonly used to identify the most significant predictors while reducing complexity (Sigurdson & Chan, 2021). Studies from MDPI and other journals highlight that not all variables are equal when looking at injury prediction. Smaller divisions of very relevant features can produce similar or even superior predictive performance to models that comparably use all variables (Cintia et al., 2021). This journal is important as it shows that efficient data collection and model implementation is especially important for practical applications.

Research shows that key predictors of injury include training load, fatigue levels, mechanical inefficiencies, and environmental conditions (Yuan et al., 2025). These findings back up the claims that injury risk is influenced by a combination of internal and external factors, which shows the need for integrated modeling approaches.

Evaluating model performance is another priority focus in literature. Many studies show accuracy as a primary metric, there is increasing recognition that accuracy alone is not sufficient enough for injury prevention tasks (Amendolara et al., 2023). Many studies emphasize the importance of recall, especially being that injury events are significantly more rare than non-injury events (Dandrieux et al., 2025). Thus, models with high accuracy may still perform poorly in identifying actual injuries. Research shows that recall is critical, as failing to identify injuries can leave significant consequences for an athlete's health and a team's performance (Leckey, 2025).

More recent studies have explored machine learning models being applied in real-world athletic environments. Research published in BMJ Open Sport and Exercise Medicine demonstrates how machine learning can be used to provide injury risk assessments on the daily and allows for coaches and medical staff to make informed decisions regarding training and recovery (Dandrieux et al., 2025). Studies that involved elite athletes and large-scale datasets have shown machine learning models can be integrated with wearable technology and athlete monitoring systems to provide information and insight into performance and injury risk (Jauhiainen, 2022). Similar work has looked at how intelligent systems can be used to enhance injury prevention and athlete monitoring (Xu & Tang, 2021).

Literature also shows several limitations, despite recent technological advancements. Many studies focus on singular sports, or restricting variables which restrict applicability of findings (Chu et al., 2022). Studies also can only evaluate only one or two machine learning models, limiting the ability to compare performance across different techniques (Amendolara et al.,

2023). A lack of comprehensive feature selection is another often seen limitation, with many not providing a broader analysis of different models and how their feature sets perform in comparison to others (Sigurdson & Chan, 2021).

Prior research has made great contributions to the field of sports injury prediction. This study extends upon existing work in several facets. Unlike many existing studies which focus on limited models or variability, this research compared multiple machine learning techniques across different groupings of features, which induces 10, 18, and 25 variables. This then allows for a more comprehensive evaluation of how feature selection impacts predictive performance.

In addition to these findings, several recent studies have further looked into the role of machine learning in injury prevention and athlete monitoring. Research has shown the importance of integrating psychological and biomechanical data to improve predictive accuracy and support injury risk assessment (Robles-Palazón et al., 2023). Other studies emphasize the role of artificial intelligence in finding the best training and rehab strategies, along with enhancing injury prevention methods (Maghsoomi et al., 2025). Other works have looked into intelligence systems and wearable technologies to monitor athlete performance and reduce injury risk (Xu & Tang, 2021).

This study in total uses over 20 academic sources, providing a broader understanding of sports injury occurring and prevention. Combining literature with modeling, this research offers a stronger framework for understanding injury risk. Along with this, this study places a strong emphasis on recall as a primary evaluation metric, aligning model assessment with real world priorities in injury prevention. The focus ensures that the models are effective both statistically, but also will be meaningful in identifying high risk athletes.

Overall, this study contributes to the existing literature about sports injuries, but this provides a more comprehensive, data driven, applicable approach to injury prediction. Hoping to merge the gap between research and real world implementation to allow athletes to be caught at risk of injuries before they happen.

3. Data Preparation

3.1 Dataset Description

The dataset used in this study consists of 10,820 observations in the training set and 1,909 observations in the test set, resulting in a large-scale dataset suitable for supervised machine learning analysis. Each observation represents an individual athlete instance and contains a comprehensive set of measurements related to physiological condition, biomechanical performance, training workload, environmental exposure, and demographic characteristics. The primary objective of the dataset is to support the prediction of injury occurrence based on patterns identified within these variables.

This dataset is structured for classification modeling, where the goal is to predict a binary outcome using multiple predictor variables. In total, the dataset contains 25 predictor variables that capture a wide range of influences on athletic performance and injury risk. These variables allow for a multidimensional analysis of injury probability, reflecting the fact that injuries are not

caused by a single factor but instead result from the interaction of multiple physiological, mechanical, and environmental conditions.

Each row in the dataset represents a unique athlete instance, which includes both internal factors (such as heart rate and fatigue) and external factors (such as temperature and playing surface). This structure allows machine learning models to identify complex patterns across different domains of athlete performance. The relatively large sample size improves the statistical reliability of the findings and reduces the likelihood of overfitting when training predictive models.

The dataset was originally provided in a structured and pre-processed format, meaning that it had already undergone initial validation before being made available for analysis. Table 1 (shown below) shows a partial display of the dataset, showcasing the structure of the observations and the range of the variables used in this study. This significantly reduces the need for manual preprocessing and ensures that the dataset is suitable for direct application in machine learning algorithms. The separation of the dataset into training and test subsets also supports proper model evaluation by allowing performance to be measured on unseen data, which is critical for assessing generalizability.

Overall, the dataset provides a strong foundation for predictive modeling due to its size, structure, and diversity of variables, all of which contribute to its suitability for analyzing injury risk in athletes. The use of large and diverse datasets has been shown to improve the effectiveness of machine learning models in injury prediction tasks (Genoud, 2024).

Figure 1: Partial Dataset Display

athlete_id	heart_ra	body_te	hydration_	sleep_qua	recovery	stress_le	muscle	joint_ang	gait_spe	cadence	step_co	jump	ground_re	range_	ambient_t	humidity	altitude	playin	training	training	trainin	fatigue_	injury_	sessio	sport_ty	gender	age	bmi
1	88.827	37.049	76.4815	5.46637	77.9	0.37583	10	126.48	2.654	50	4961	0.5	1859	113.4	17.6032	47.315	34.8536	0	5.678	79.11	548.4	45.029	0	1	Basketb	Female	18	25.1
1	40	36.93	72.3639	6.89909	74.845	0.27916	310.3	157.63	3.431	89.033	9006	0.3	1852.052	146.2	25.1071	72.398	241.114	1		117	538	42.255	1	2	Basketb	Female	18	25.1
1	76.175	36.335	100	9.57415	83.538	0.33406	158.7	81.766	0.8	73.088	6775	0.9	1654.116	120.7	27.895	64.386	236.488	1	6.643	96.35	647.8	61.87	0	3	Basketb	Female	18	25.1
1	100.54	37.244	86.1725	7.6143	79.967	0.21877	107.1	113.02	1.714	50	6579	0.5	1727.371	142.7	27.8864	46.833	290.514	0	6.194	97.62	336.6	50.269	0	4	Basketb	Female	18	25.1

3.2 Target Variable

The target variable for this study is injury_occurred, which represents whether an athlete experienced an injury during a given observation period. This variable is binary in nature and is encoded as follows:

- 0 = No injury occurred
- 1 = Injury occurred

This binary classification structure defines the problem as a supervised learning classification task. The objective of all predictive models developed in this study is to accurately classify whether an injury is likely to occur based on the provided predictor variables.

The importance of this target variable lies in its real-world implications. Injury prediction is a critical problem in sports science, as injuries can significantly impact athlete performance, career

longevity, and team outcomes. As a result, models that can accurately predict injury occurrence provide valuable insights for coaches, trainers, and medical staff.

In evaluating model performance, accuracy alone is not sufficient due to the potential consequences of misclassification. In particular, false negatives (failing to predict an injury when one occurs) are more critical than false positives. Therefore, additional evaluation metrics such as precision, recall, and F1-score are necessary to fully assess model effectiveness. Recall is especially important in this context because it measures the model's ability to correctly identify injury cases.

Although the dataset does not explicitly indicate severe class imbalance, the use of multiple evaluation metrics ensures that performance is evaluated in a balanced and meaningful way. This helps prevent misleading conclusions that could arise from relying solely on accuracy.

3.3 Features

The dataset contains 25 predictor variables that represent multiple dimensions of athlete health, performance, and environment. These variables are grouped into five major categories: physiological, biomechanical, training-related, environmental, and demographic features.

Physiological Features

Physiological variables describe the internal biological and health-related state of the athlete. These include heart_rate, body_temperature, hydration_level, sleep_quality, stress_level, and bmi. These features are important indicators of recovery, fatigue, and overall physical condition. For example, elevated heart rate may indicate physical stress or insufficient recovery, while poor sleep quality may reduce the body's ability to repair and regenerate. High stress levels can also contribute to fatigue accumulation, increasing the likelihood of injury over time.

Biomechanical Features

Biomechanical variables describe movement patterns and physical mechanics during athletic activity. These include joint_angles, gait_speed, cadence, step_count, jump_height, ground_reaction_force, and range_of_motion. These variables are critical for understanding how an athlete moves and how physical forces are distributed across the body. Abnormal movement patterns, such as inefficient joint angles or excessive ground reaction force, can place strain on muscles and joints, increasing injury risk. Similarly, limited range of motion may restrict movement efficiency and contribute to compensatory movement patterns that increase stress on the body.

Training and Workload Features

Training-related variables include training_intensity, training_duration, training_load, and fatigue_index. These variables represent the volume and intensity of physical activity performed by the athlete. Training load is particularly important, as excessive load without adequate

recovery is a well-established risk factor for injury. Fatigue index captures accumulated physical strain, which can reduce coordination, reaction time, and movement control. Together, these variables provide insight into how workload contributes to injury risk.

Environmental Features

Environmental variables include ambient_temperature, humidity, altitude, and playing_surface. These variables represent external conditions that may influence athlete performance and physical stress. High temperatures and humidity can lead to dehydration and increased fatigue, while altitude can affect oxygen availability and endurance. The playing surface also plays a significant role, as different surfaces can alter impact forces and joint stress during movement.

Demographic Features

Demographic variables include age, gender, and sport_type. These variables provide contextual information about the athlete and help account for differences in physical development, physiological capacity, and sport-specific demands. For example, older athletes may have slower recovery rates, while different sports place varying levels of stress on the body.

Overall, these 25 features collectively provide a comprehensive representation of the factors influencing injury risk, enabling robust predictive modeling.

3.4 Preliminary Analysis

The preliminary analysis focused on evaluating how different feature set sizes impact model performance across multiple machine learning algorithms, rather than examining individual variables in isolation. The analysis compared model behavior using small, medium, and large feature sets, providing insight into how increasing complexity affects predictive accuracy, recall, and error rates.

When transitioning from a small to a medium feature set, most models demonstrated slight improvements in performance. Support Vector Machines, Decision Trees, and Neural Networks all showed increases in accuracy and macro recall, along with reductions in error rate. For example, SVM exhibited a measurable improvement in both accuracy and recall, indicating that the additional features provided useful information that enhanced model learning. Decision Trees also showed improved performance, particularly in recall, suggesting better identification of injury cases. Neural Networks benefited modestly as well, reflecting improved learning from a richer feature set.

These improvements suggest that the initial increase in features helped models capture additional meaningful relationships between predictors and injury occurrence. The medium feature set appears to provide an optimal balance between information richness and noise reduction.

However, when expanding from a medium to a large feature set, a consistent decline in model performance was observed across all algorithms. Accuracy decreased, macro recall dropped, and

error rates increased. This pattern suggests that the inclusion of additional features beyond a certain threshold introduces noise and reduces model generalization ability. Instead of improving performance, the larger feature set likely caused overfitting or diluted the influence of the most important variables.

Naive Bayes showed the most significant decline in performance, indicating sensitivity to irrelevant or redundant features. Neural Networks also experienced reduced performance, likely due to increased model complexity and difficulty in learning meaningful patterns from noisy data. Decision Trees remained relatively stable but did not significantly benefit from additional features, while SVM showed a decline after initial improvements, suggesting an optimal feature range exists for best performance.

Overall, the analysis demonstrates that increasing the number of features does not always lead to improved predictive performance. Instead, there is an optimal range in which models perform best, with medium feature sets providing the most consistent and balanced results across all algorithms.

3.5 Data Cleaning

The data cleaning process for this study was conducted prior to analysis using the Ohio Supercomputer Center in conjunction with MATLAB. These tools provided a high-performance computing environment that enabled efficient and systematic preprocessing of the dataset. As a result, the dataset used in this study was fully pre-processed and validated before being imported for modeling.

Because data cleaning was completed externally, no additional manual cleaning procedures were required in this study. The dataset was considered complete, consistent, and free from major errors or inconsistencies at the time of analysis. This allowed the research to focus entirely on model development, feature interpretation, and performance evaluation rather than raw data correction.

Missing values were addressed during the preprocessing stage in MATLAB. Depending on the nature of the missing data, appropriate methods such as imputation or removal were applied to ensure dataset completeness. As a result, no missing values were present in the final dataset used for analysis. This ensured that all machine learning models were trained on complete observations without bias introduced by missing data handling during modeling.

Outlier detection and treatment were also conducted during preprocessing. Extreme or unrealistic values were identified and corrected or removed to ensure that all variables remained within acceptable and interpretable ranges. This step is particularly important in datasets involving physiological and biomechanical data, where abnormal values can significantly distort model training and predictions.

Normalization and scaling were applied as part of the preprocessing pipeline to ensure consistency across variables with different units and magnitudes. Since the dataset includes

variables such as heart rate, training load, and environmental measures, scaling ensures that no single variable dominates model training due to its numerical range. This is especially important for algorithms such as Support Vector Machines and Neural Networks, which are sensitive to feature scaling.

The use of the Ohio Supercomputer Center and MATLAB ensured that preprocessing was conducted using reliable, efficient, and scalable computational methods. This not only improved the quality of the dataset but also ensured reproducibility and consistency in the data preparation process.

Overall, the dataset required no additional cleaning within this study due to the thorough preprocessing completed beforehand. This allowed for a streamlined transition into model development while maintaining a high level of confidence in data quality and integrity.

4. Methodology

4.1 Models used

To help dive into injury occurrence prediction, machine learning uses multiple different classification models. These models are implemented and evaluated, and because the target variable is binary being injured or not, classification algorithms are the best for this type of analysis. All models were selected based on their ability to handle the data and its structure, capture the relationship between variables, and provide accurate predictions in the given context.

Naive Bayes is a probability outcome based algorithm. It assumes all predictor variables are independent of each other given the outcome. This then allows for the model to simplify its computation and be efficient in training (Sigurdson & Chan, 2021). Naive Bayes was especially efficient in this study due to its simplicity, speed, and effectiveness when working with larger variables containing datasets. Even with the strong independence assumption, studies in the past show that

Naive Bayes can perform in complex classification datasets (Amendolara et al., 2023). In this given prediction, the model estimates the probability that an injury will occur based on features such as fatigue, workload, and many other predictors. This allows Naive Bayes to be a productive model to compare against other more complex models.

Logistic regression is a classification model that predicts the probability of a binary outcome by using the relationship between predictors and the odds of the target variable. It will make probability outputs between 0 and 1, being used for classification cases. Logistic regression is widely used in sports injury to identify relationships within training load, fatigue, and injury risk (Cintia et al., 2021). Logistic regression here served as a benchmark, allowing for interpretation of how the variables correlate to injury occurrence. Logistic regression has the ability to handle multiple predictors and differentiate between injury and noninjury cases.

Decision Trees are a classification model that splits the data into branches based on feature values, forming a tree-like structure that leads to the outcomes. The splits are determined by selecting the feature that best separates the data into different classes. Decision trees are used due to the ability to capture nonlinear relationships and interactions between variables. Injury risk is influenced by combinations of factors rather than single variables, which allowed decision trees

to be useful in this particular case (Yuan et al., 2025). Trees can model interactions well while remaining interpretable, allowing visualization of how. These methods are especially useful in settings when understanding the model decision is important.

Support Vector Machines are supervised learning models that learn from labeled data. They classify information by understanding the hyperplane between the classes. SVM models are particularly effective in cases where the relationship between variables is complex and not easily separable. In this case, it is well used due to its performance in high dimensional classification problems. SVM models are effective in injury prediction due to their ability to handle relationships and complex interactions among variables (Chu et al., 2022; Robles-Palazón et al, 2023). Given the large feature selection in this dataset, SVM produces a strong method for patterns associated with injury occurring and classification performance.

Neural Networks are models composed of layers of nodes that process data and learn patterns through weighted connections. Neural Networks are capable of capturing nonlinear relationships between variables. Research has shown the effectiveness of neural networks in modeling injury risk due to their ability to process high dimensional data (Raju et al., 2026). Their ability to model complex and high dimensional data makes them useful in this study. Injury occurrence is influenced by multiple pieces, including physiological, biomechanical, and environmental factors (Amendolara et al., 2023). Although Neural Networks are not as interpretable as some other models, they provide strong predictive performance when trained on larger datasets like the one used in this study.

The use of multiple machine learning models in this study allows for a more comprehensive comparison of different approaches to injury prediction. All models have their values, with Naive Bayes and Logistic Regression being simpler, and giving easier interpretability and baseline comparisons. The complex models like Trees, SVM, and Neural Networks offer higher ability to capture nonlinear and high dimensional relationships (Leckey et al, 2025). Injury prevention being a classification problem that happens in the real world frequently, it is important to evaluate multiple models and find effective approaches. All models have pros and cons, with trade offs for interpretability and predictive performance. Combining multiple modeling approaches has been seen in prior research to provide a more well rounded understanding of injury risk compared to simply one model (Dandrieux et al., 2025).

4.2 Feature Selection

Feature Selection is a major step in machine learning. It prioritizes improving model performance, reducing overfitting and enhancing interpretability by getting the most relevant predictors. Within the context of injury prediction, the variables in this specific case consisted of physiological, training, performance, environmental, and recovery variables. Not all of these features contribute equally when looking at predicting injury occurrence, which is why feature selection becomes very important. Past research has shown that selecting the most important and relevant features can improve accuracy and reduce redundancy or irrelevant variables (Sigurdson & Chan, 2021; Cintia et al., 2021).

To make sure of a data driven approach, feature selection here was done by using multiple statistical and machine learning techniques. Feature importance was evaluated using Minimum

Redundancy Maximum Relevance (MRMR), Analysis of Variance (ANOVA), Chi-Square (Chi2), and Kruskal Wallis. All of these methods help to assess feature relevance from a valuable perspective, and allow for a comprehensive evaluation of variables importance.

MRMR is designed to highlight the features that are most relevant to the target variables while minimizing redundancy among the predictors. This makes it particularly effective for reducing multicollinearity in high dimensional datasets (Sigurdson & Chan, 2021). ANOVA evaluates the statistical significance of differences between groups, finding the variables that most strongly differentiate between injury and no injury occurrence. Chi-Square testing measures the dependence between the categorical variables and the targeted outcome, while Kruskal Wallis provides a no parameter alternative test for comparing the distribution across groups. This makes it viable for variables that do not follow normal distribution.

To create the final feature selection rankings, the scores across all four methods were combined and calculated by taking the average of the scores for each variable. This approach reduces the bias that could come with a particular method and scores them fairly on all aspects, with final rankings reflecting a balanced approach. The results then were used to construct three feature sets for analysis, the large set containing all 25 variables, the medium containing 18 of those, and the small containing 10 variables. Figure 2 (shown below) shows the rankings of the 25 features based on the combined results after running the 4 methods. Showing that recovery score, sleep quality, and fatigue index were identified as the most influential predictors.

Figure 2: Feature Rankings

Feature Name	MRMR	Anova	Chi2	Kruskal Wallis	Average Score
Recovery Score	1	1	1	1	1
Sleep_Quality	5	2	2	2	2.75
Fatigue_Index	3	3	3	3	3
training_load	10	4	4	4	5.5
Gender	2	10	6	9	6.75
hydration_level	12	5	12	5	8.5
Playing_Surface	4	13	9	13	9.75
altitude	13	7	13	7	10
stress_level	18	9	5	10	10.5
joint_angles	7	11	14	12	11
training_intensity	26	8	8	8	12.5
body_temperature	21	6	19	6	13
gait_speed	15	12	16	11	13.5
sport_type	16	15	20	15	16.5
muscle_activity	17	17	17	16	16.75
training_duration	25	16	11	17	17.25
age	11	20	22	19	18
heart_rate	19	22	10	21	18
humidity	8	19	25	20	18.25
ambient_temperature	9	23	18	23	18.25
range_of_motion	24	21	7	22	18.5
bmi	27	14	24	14	19.75
athlete_id	28	18	21	18	21.25
Session_Id	6	27	28	26	21.75
jump_height	20	26	15	27	22
step_count	22	25	23	25	23.75
cadence	14	28	27	28	24.25
ground_reaction_force	23	24	26	24	24.25

The large feature set includes all available predictors from all categories and provides the most comprehensive representation of injury risk factors. The medium set containing 18 variables consists of the highest ranked predictors after removing the least influential or redundant features. This allows a good balance between information and model simplicity. The small

feature set included only the most relevant and significant predictors across all ranking methods. These focused on key factors like fatigue, workload, stress and environmental conditions. All of these have been consistently identified in past research as major contributors to injury risk (Yuan et al., 2025).

To evaluate the impact of feature selection on model performance, all models were trained and tested on all of the feature sets. This allowed for a direct comparison of how reducing the number of predictors affects the key performance metrics, including accuracy, recall, and precision. After conducting the preliminary analysis, it was seen that increasing the number of features can slightly improve performance, but only to a certain point. This comes as the large feature set was worse as it came to accuracy compared to the medium training sets, as the large feature set increases noise and reduces model generalization.

The medium feature set demonstrated the most consistent and balanced performance across the models. This shows that there is an optimal range of predictors for injury prediction. This comes to align with prior research, which emphasizes that more features do not always lead to better performance, and that carefully selecting the subsets of variables can produce better results (Cintia et al., 2021; Amendolara et al., 2023).

Overall, the feature selection process provides a structured and evidence based method for identifying the most important predictors of injury occurrence. Combining multiple ranking techniques and evaluating different feature groups, this study not only improves predictive performance but also enhances understanding of the key variables that contribute to injury risk.

4.3 Design of Experiments

The experimental design for this study was structured to ensure reliable and unbiased evaluation of model performance across different feature sets and machine learning algorithms. The objective being injury occurrence, which is a binary classification problem, means the models were evaluated on unseen data to truly assess their ability to generalize beyond the training set.

The data was divided into two subsets, a training set and a testing set. The training set contained 10,820 observations, while the testing set contained 1,909 observations. The training data was used to develop and fit each machine learning model, allowing the algorithms to learn patterns and relationships between the predictor variables and injury occurrence. The testing set, which was unused during training, was then used to evaluate model performance. This is critical in machine learning, as it helps to prevent overfitting and provides a real measure of how well the model performs on new data. Evaluating models on the testing set ensures that the results reflect true predictive capability rather than memorizing the training data.

To improve reliability and model strength, each model was run 5 times for each feature set grouping. This resulted in multiple performance measurements for each model-feature combination, allowing a more stable and consistent evaluation.

These repeated runs are important because many machine learning algorithms involve elements of randomness. With this, model performance can vary slightly, and conducting multiple runs and analyzing average performance allows this study to reduce the impact of random variation

and ensures that the results are not dependent on a single execution. This allows for more accurate comparison and improved statistical reliability between models. It also provides a stronger foundation for identifying consistent performance patterns across the feature sets.

All machine learning models were trained and evaluated across all feature sets. This created a comprehensive comparison that allowed for analysis of both model performance and the impact of feature selection. By combining multiple models, multiple feature sets, and repeated runs, this experimental design ensures a strong and systematic evolution of injury prediction. This approach allows better comparisons and supports identification of the most effective models and feature combinations for injury occurrence prediction.

5. Results

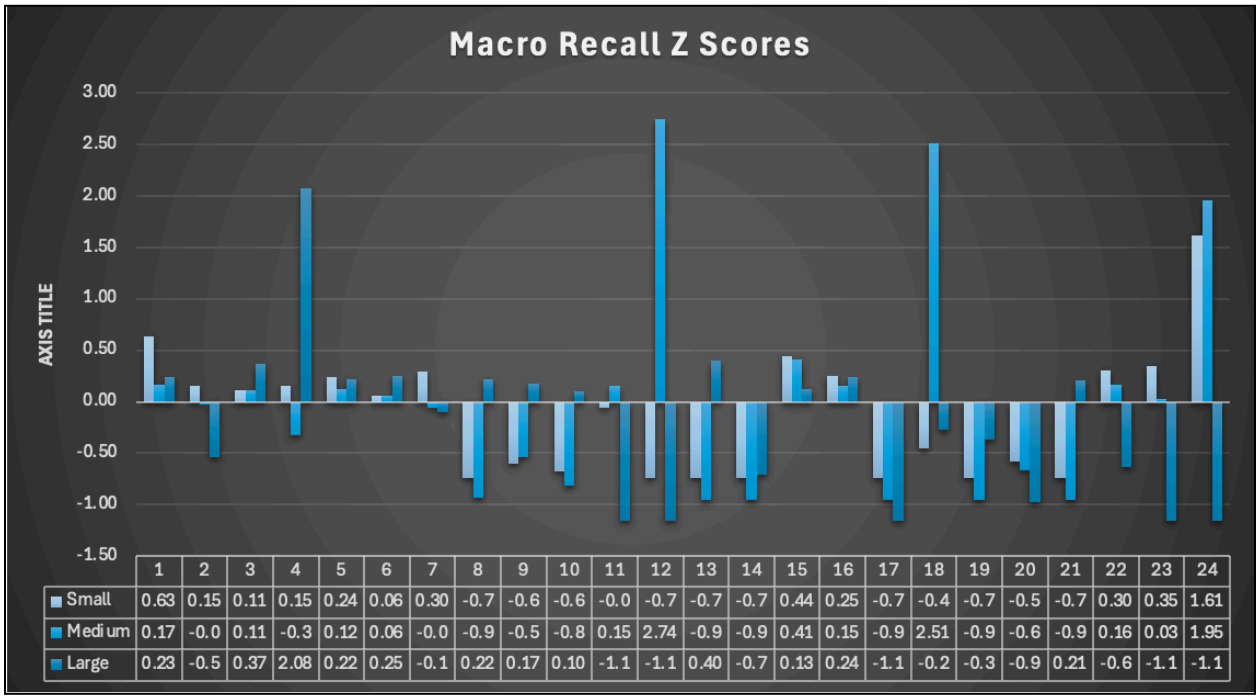
The results of this study focus on evaluating the performance of multiple machine learning models across the different feature set sizes and identifying which models were most effective for predicting injury occurrence. The model performance was assessed using a weighted score approach that included macro recall, accuracy, macro precision, and training time. Z scores were calculated and used then to standardize the metrics, allowing for a fair and accurate comparison across the models with different scales. Macro recall was weighted to 50%, while accuracy and precision were assigned a 20% weight, and training time at 10% to reflect the predictive effectiveness and efficiency.

Z Score Formula :	Overall Score Formula :
=(D10-AVERAGE($D$10:$D$33))/STDEV.S($D$10:$D$33)	=(0.5*I10)+(0.2*H10)+(0.2*J10)-(0.1*K10)

5.1 Model Performance Across Feature Sets

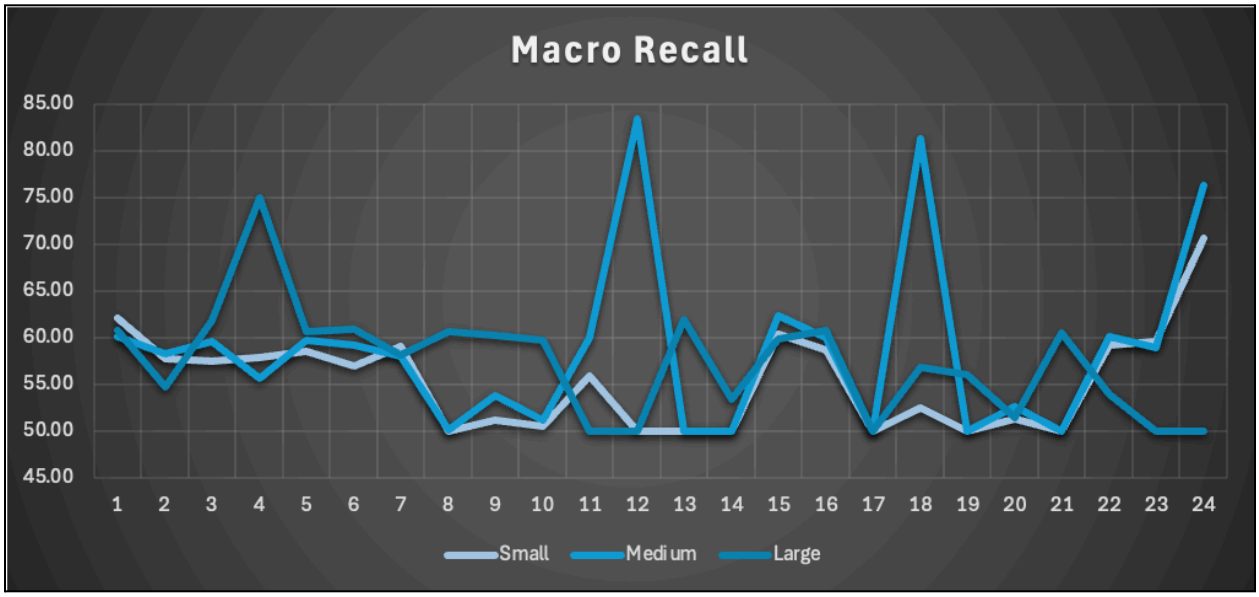
This analysis compared model performance using three feature sets, the large consisting of 25 variables, medium consisting of 18, and small using 10. Figure 4 which can be found below showcases the compassion of macro recall across the models in the small, medium, and large feature sets. These consistent findings with the preliminary analysis show that performance improved when moving from the small to the medium feature set, but then declined when expanding to the large set. Figure 3 illustrates the macro recall z scores across all models in all feature sets. This figure highlights the superior consistency of the medium feature set as it pertains to others.

Figure 3: Macro Recall Z Scores



	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24
Small	0.63	0.15	0.11	0.15	0.24	0.06	0.30	-0.7	-0.6	-0.6	-0.0	-0.7	-0.7	-0.7	0.44	0.25	-0.7	-0.4	-0.7	-0.5	-0.7	0.30	0.35	1.61
Medium	0.17	-0.0	0.11	-0.3	0.12	0.06	-0.0	-0.9	-0.5	-0.8	0.15	2.74	-0.9	-0.9	0.41	0.15	-0.9	2.51	-0.9	-0.6	-0.9	0.16	0.03	1.95
Large	0.23	-0.5	0.37	2.08	0.22	0.25	-0.1	0.22	0.17	0.10	-1.1	-1.1	0.40	-0.7	0.13	0.24	-1.1	-0.2	-0.3	-0.9	0.21	-0.6	-1.1	-1.1

Figure 4: Macro Recall



The small feature set which only included the 10 predictors that were most relevant, produced on average lower recall and overall performance across models. This suggests that these variables are important, but do not capture everything and full complexity of an injury occurring. As a result of this, models that were trained only on the small feature set are more likely to miss an injury, due to underfitting with insufficient information.

The medium feature set included all 10 of the small feature sets variables as well as the next 8 labeled as most relevant. This demonstrated a good balanced performance across the models.

Many models showed improvement in both recall and accuracy compared to small feature set. This indicated that the 8 additional variables provided meaning and necessary information without adding excess noise. This is also seen in prior research findings that emphasized the importance of selecting an optimal amount of predictors rather than maximizing the total number of features in machine learning algorithms.

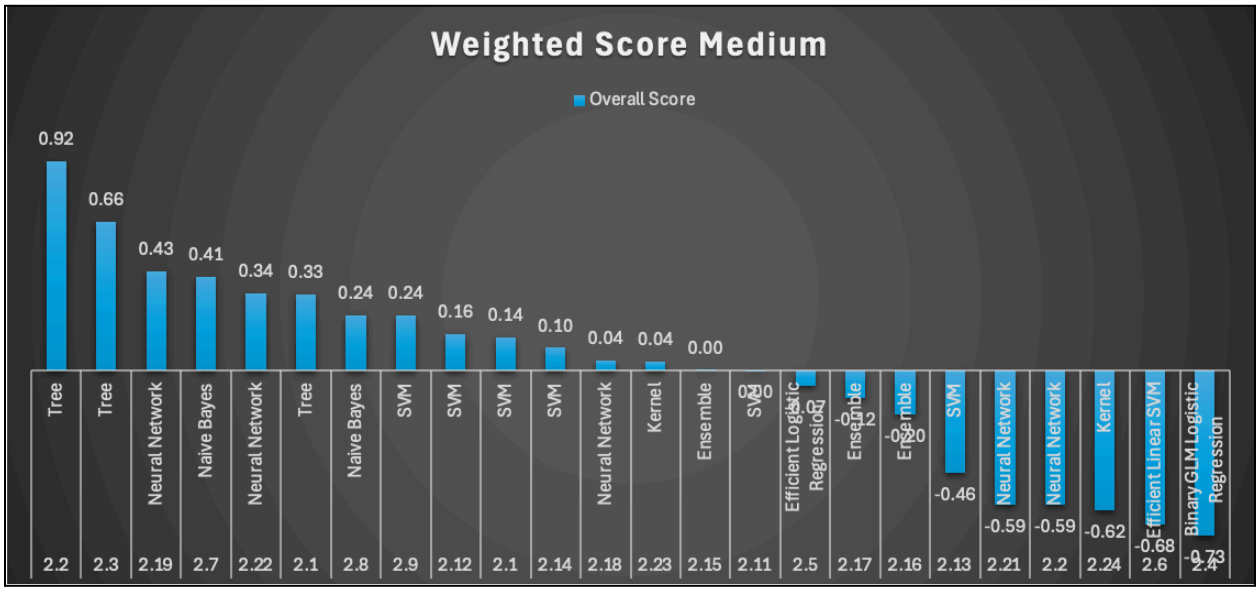
In contrast, the large feature set which contained all 25 variables, showed a decline in performance in several models as compared to the medium set. Although it contains the most comprehensive and broadest set of predictors, due to its inclusion of irrelevant or redundant variables added noise and reduced the models generalization ability. This lined up with existing research that shows how models with excessive features can suffer with overfitting problems and reduced predictive accuracy. Similar findings have been seen in studies that emphasize the importance of optimal feature selection in improving predictive performance (Guelmami et al., 2023).

5.2 Individual Model Comparison

Figure 5: Machine Learning Model List

Model Number	Model Type
2.1	Tree
2.1	SVM
2.11	SVM
2.12	SVM
2.13	SVM
2.14	SVM
2.15	Ensemble
2.16	Ensemble
2.17	Ensemble
2.18	Neural Network
2.19	Neural Network
2.2	Tree
2.2	Neural Network
2.21	Neural Network
2.22	Neural Network
2.23	Kernel
2.24	Kernel
2.3	Tree
2.4	Binary GLM Logistic Regression
2.5	Efficient Logistic Regression
2.6	Efficient Linear SVM
2.7	Naive Bayes
2.8	Naive Bayes
2.9	SVM

Figure 6: Weighted Score Medium



To look at the best individual model performance, all the individual model runs were evaluated using a weighted z score approach. Recall was assigned the highest weight due to the importance it has in correctly identifying injury cases, as well as its ability to penalize missing injury occurrence. Accuracy and precision followed this and also factored into the weighted score at 20% each to ensure the models overall reliability. Training time was the last piece and was included as a factoid to account for the efficiency of the model but only weighted at 10%.

Using the medium feature set, individual models were ranked based on the overall weighted scores. Figure 5 (shown above) gives a list of all 24 models that were evaluated, along with their model type. Figure 6, which can be seen below that, showcased the weighted z-score rankings for all models in the medium feature set. Among the 24 models evaluated, Model 2.2 a Decision Tree came back with the highest weighted score and was then identified as the best performing model in this case. This model performed exceptionally well on recall, getting 93.43% which was the highest among all models, showcasing an extremely strong ability to identify injury cases. Even with low precision and a training time on the higher end, it still achieved the highest score due to an above average accuracy and an extremely high recall score.

The second best model was Model 2.3, also a Decision Tree. This demonstrated strong recall as well at 81.32% and further backed up evidence that the tree based models were some of the strongest in this study.

5.3 Recall Analysis

In this study, recall was the most important evaluation metric due to its critical nature of correctly identifying the injury cases and penalizing those that are missed. Models that have a higher recall are the most effective in detecting injury cases and reduce the likelihood of false negatives or missed injuries. Both Model 2.2 and 2.3 achieved the highest recall in the medium feature set among the models, and ended up with the highest weighted score making them the most effective models. These findings showcase the strength of the decision tree models in capturing patterns within injury occurrence in machine learning.

These models do have some flaws though, as they both experienced lower precision, indicating a higher rate of false positives. This indicates a common challenge that can be associated with classification problems, as improving recall can come with potential weakness in precision.

5.4 Model Performance Trends

Model performance within the medium feature set showed some trends across model types. Decision tree models consistently outperform other models, especially in recall, showcasing a strong ability and performance in this study. Neural Networks and Naive Bayes demonstrated moderate performance, having balanced but overall lower scores compared to trees with the medium feature set. Support Vector Machines showcased sporadic performance, some models perform well but the overall consistency is lacking.

Logistic Regression models performed poorly as it pertained to the medium feature set. These models experienced low recall values, making the overall weighted scores lower than other models. Model 2.5 a Binary GLM Logistic regression achieved the lowest overall score, showing an inability to identify injury cases in this case. With these findings, logistic regression may be less suitable for complex and high dimensional injury prediction studies compared to the other machine learning styles.

5.5 Key Takeaways

The results of this study highlight several important findings. Firstly, the medium feature set provided the most reliable and consistent performance. This makes it the most appropriate for evaluating these models. As a result of this, the individual model rankings were taken from the medium feature set as they were considered the most accurate representation of model efficiency and effectiveness.

Second, decision tree models came to be the top performing approach, partially due to their ability to score high on recall and capture the relationship between complex variables. Recall is also shown to remain a critical metric in injury prediction, as models with high accuracy may still fail to showcase missed injury events occurring. Lastly, the results emphasize the importance of evaluating models using multiple metrics rather than solely relying on accuracy. Using and prioritizing recall and other performance metrics allow this study to provide a more realistic and meaningful assessment of model effectiveness in injury prediction.

6. Performance Assessments

Performance Assessment provides a deeper evaluation of the models and their effectiveness in predicting injury occurrence. In the context of this case, failing to predict an injury (false negatives) has a significant real world impact which allows us to place a heavy emphasis on recall. With this, recall served as the primary measure of model performance in this study.

6.1 Recall and Injury Detection

Across all three of the feature sets, the models that achieved the highest recall were most effective at identifying injury cases. As mentioned prior, Models 2.2 and 2.3 attained the highest score in the medium feature set, largely due to their exceptional recall scores. This allows for us to know the effectiveness of tree based models for this study and their ability to accurately

predict injury occurrence. These findings come to be consistent with research that highlighted the importance of recall in identifying injury cases within datasets (Zhao, 2024).

When looking at all of the feature sets, similar patterns were observed. The large feature set saw Model 2.17 Ensemble have the highest recall at 75% showcasing its ability to make injury detection. The small feature set had the highest recall be from Model 2.9 and SVM, this time it ranked 9th in overall models for the feature set, still helping to reinforce the case that recall plays a significant factor. These findings help to suggest that the models with higher recall consistently outperform others when identifying injury cases, making them the most suitable for real world scenarios.

6.2 Tradeoffs in Recall and Accuracy

Something to take note of is the tradeoff between recall and other metrics such as accuracy and precision. Models with the better recall sometimes shower lower scores on precision, meaning they had more false positives. An example of this is Model 2.2 in the medium feature set. This model had the highest recall but significantly lacked precision, indicating a strong injury detection but reduced classification accuracy. In contrast to this, models like logistic regression had higher accuracy but were often lower with recall, making them less effective for injury identification. These results help to solidify the case that missing an injury is more important in cases than incorrectly predicting an injury.

6.3 Model Comparison

Decision tree models consistently achieved high recall scores among all the feature sets. This makes these models the best for injury detection. Ensemble models showed good performance but were not as consistent, and neural networks and SVM models were more balanced in their performance.

6.4 Overall Insights

Overall insights showed that the medium feature set produced the most consistent and reliable results, furthering the fact that it is the best set to evaluate. These results overall show that :

- Recall is the most important metric for injury prediction
- Decision tree models had the best performance
- There are tradeoffs between recall and precision
- The medium feature set had the best balance between performance and generalization

7. Conclusion

This study shows the ability of machine learning techniques for predicting sports injury occurrence in a data driven way. By looking at 12,000 observations taken from athletes and evaluating multiple different models across multiple different feature sets, the research here showcases how different training related factors can contribute to injury risk.

One of the key components of this study was feature selection, the medium feature set produced the most consistent and balanced results. The medium feature set produced better results showcasing that it had the best number of predictors for this type of classification problem. Too few variables limited the models performance, and too many features added noise and reduced the generalization.

The decision tree models came to the top as the most effective, especially in the medium feature set. This came about as they had high recall and ability to capture the complex relationships. Model 2.2 achieved the highest overall weighted score, proving that it has a high ability to accurately identify injury cases. This also came with one of the main tradeoffs of the study, as missing an injury has more consequences than overpredicting one happening as that is how it is viewed in the real world.

The results of this study also showcase the difference between recall and precision. Higher recall models may produce some false positives, they are more beneficial in this particular case than a precision model which may produce missed injuries which can be catastrophic. This study overall contributes to sports analytics by combining multiple different machine learning models, feature selection methods, and evaluation metrics. It enhances upon prior research and compares a variety of models and feature sets.

In the future, integrating these models into technology that can be worn by athletes in devices such as WHOOP, Oura Ring, smart watchers and more can enhance real-time injury prediction. The monitoring of recovery, workload, and physiological data offers strong potential for improving detection and supporting data driven decision making.

In conclusion, machine learning being used with proper feature selection and evaluation metrics can significantly improve injury prediction. The findings in this study hope to provide practical value for enhancing and supporting athlete health, financial impact, and long term athletic performance.

References

Amendolara, A., Pfister, D., Settelmayer, M., Shah, M., Wu, V., Donnelly, S., … Bills, K. (2023). An Overview of Machine Learning Applications in Sports Injury Prediction. *Cureus*. https://doi.org/10.7759/cureus.46170

Chu, Y., Knell, G., Brayton, R. P., Burkhart, S. O., Jiang, X., & Shams, S. (2022). Machine learning to predict sports-related concussion recovery using clinical data. *Annals of Physical and Rehabilitation Medicine*, *65*(4), 101626. https://doi.org/10.1016/j.rehab.2021.101626

Dandrieux, P.-E., Navarro, L., Blanco, D., Ruffault, A., Ley, C., Bruneau, A., … Edouard, P. (2025). Association between the use of daily injury risk estimation feedback (I-REF) based on machine learning techniques and injuries in athletics (track and field): results of a prospective cohort study over an athletics season. *BMJ Open Sport & Exercise Medicine*, *11*(1), e002331. https://doi.org/10.1136/bmjsem-2024-002331

Gai, X. (2025). Discover Artificial Intelligence Application of flexible sensor multimodal data fusion system based on artificial synapse and machine learning in athletic injury prevention and health monitoring. *Discover Artificial Intelligence*, *5*, 31. https://doi.org/10.1007/s44163-025-00254-4

Genoud, M. (2024). *Statistics and Machine Learning in Sports Injury Prevention*. Retrieved from https://bodai.unibs.it/bdsports/wp-content/uploads/sites/2/2024/07/Bachelor-thesis-Melissa-GENOUD.pdf

Guelmami, N., Fekih-Romdhane, F., Mechraoui, O., & L. Bragazzi, N. (2023). Injury Prevention, Optimized Training and Rehabilitation: How Is AI Reshaping the Field of Sports Medicine. *New Asian Journal of Medicine*, *1*(1), 30–34. https://doi.org/10.61186/najm.1.1.30

Jauhiainen, S., Kauppi, J.-P., Krosshaug, T., Bahr, R., Bartsch, J., & Äyrämö, S. (2022). Predicting ACL Injury Using Machine Learning on Data From an Extensive Screening Test Battery of 880 Female Elite Athletes. *The American Journal of Sports Medicine*, *50*(11), 2917–2924. https://doi.org/10.1177/03635465221112095

Jauhiainen, S., Kauppi, J.-P., Leppänen, M., Pasanen, K., Parkkari, J., Vasankari, T., … Äyrämö, S. (2020). New Machine Learning Approach for Detection of Injury Risk Factors in Young Team Sport Athletes. *International Journal of Sports Medicine*, *42*(02), 175–182. https://doi.org/10.1055/a-1231-5304

Leckey, C., van Dyk, N., Doherty, C., Lawlor, A., & Delahunt, E. (2024). Machine learning approaches to injury risk prediction in sport: a scoping review with evidence synthesis. *British Journal of Sports Medicine*, *59*(7), bjsports-2024-108576. https://doi.org/10.1136/bjsports-2024-108576

Maghsoomi, M., Johari, K., & Abedini, E. (2025). Artificial neural networks for prevention of sports injuries: a systematic review. *Sport Sciences for Health*. https://doi.org/10.1007/s11332-025-01504-9

Petrovsky, D. V., Pustovoyt, V. I., Nikolsky, K. S., Malsagova, K. A., Kopylov, A. T., Stepanov, A. A., … Kaysheva, A. L. (2022). Tracking Health, Performance and Recovery in Athletes Using Machine Learning. *Sports*, *10*(10), 160. https://doi.org/10.3390/sports10100160

Raju, S., Singamaneni, K. K., Hooi, L. B., Rani, K. U., & B., C. (2026). Machine learning framework for predicting athletic injuries and optimising performance. *BMC Sports Science, Medicine and Rehabilitation*. https://doi.org/10.1186/s13102-025-01502-x

Robles-Palazón, F. J., Puerta-Callejón, J. M., Gámez, J. A., De Ste Croix, M., Cejudo, A., Santonja, F., … Ayala, F. (2023). Predicting injury risk using machine learning in male youth soccer players. *Chaos, Solitons & Fractals*, *167*, 113079. https://doi.org/10.1016/j.chaos.2022.113079

Rossi, A., Pappalardo, L., & Cintia, P. (2021). A Narrative Review for a Machine Learning Application in Sports: An Example Based on Injury Forecasting in Soccer. *Sports*, *10*(1), 5. https://doi.org/10.3390/sports10010005

Sigurdson, H., & Chan, J. (2021). Machine Learning Applications to Sports Injury: A Review. *Proceedings of the 9th International Conference on Sport Sciences Research and Technology Support*. https://doi.org/10.5220/0010717100003059

Tsilimigkras, T., Kakkos, I., Matsopoulos, G., & Bogdanis, G. (2024). Enhancing Sports Injury Risk Assessment in Soccer Through Machine Learning and Training Load Analysis. Retrieved from https://pmc.ncbi.nlm.nih.gov/articles/PMC11366842/

Xu, T., & Tang, L. (2021). Adoption of Machine Learning Algorithm-Based Intelligent Basketball Training Robot in Athlete Injury Prevention. *Frontiers in Neurorobotics*, *14*. https://doi.org/10.3389/fnbot.2020.620378

Yuan, J. (2026). Machine Learning Applications in Non-Contact Lower Limb Sports Injury Prediction: A Systematic Review. *Journal of Sports Science and Medicine*, *25*(1), 172–194. Retrieved from https://www.jssm.org/jssm-25-172.xml-Fulltext

Yuan, J., Zeng, Q., Li, J., Cong, Z., & Zhang, Y. (2025). Machine learning applications in sports injury prediction: A narrative review. *Science Progress*, *108*(4), 368504251385956-368504251385956. https://doi.org/10.1177/00368504251385956

Zhao, F. (2024). The Application of Sports Biomechanics in Sports Injury Prevention and Rehabilitation. *Frontiers in Sport Research*, *6*(3). https://doi.org/10.25236/fsr.2024.060320